

Crypto Investigator

Overview and Quick Start — v1.0.0, 2026-09-04 23:45 UTC

What it does

A cryptocurrency tracing and warrant-preparation tool for law enforcement. Starting from a victim's wallet, it follows stolen or scammed funds across the public blockchain until they reach a business - usually a crypto exchange - that can be served with a subpoena or warrant for the account holder's records. It then packages everything needed to draft that legal process: the deposit addresses, funding transactions, timestamps, amounts (with approximate USD values at the transaction dates), and draft records-request language.

It does not identify people. It identifies where legal process can be served; the human behind an account comes from the exchange's KYC records via that process. It never guesses: everything that is an inference rather than an on-chain fact is labelled with a confidence level and its basis, and dead ends (mixers, privacy coins, smart contracts) are reported honestly.

How it works, at 30,000 feet

1. The investigator creates a case and enters the victim's wallet address - and, if known, the transaction hash of the specific payment to the scammer, so only that payment is followed.
2. The tool pulls public blockchain records and follows the money hop by hop (breadth-first, up to a chosen depth), recording every address and movement.
3. Each address it reaches is checked against the official OFAC sanctions list, public exchange/mixer label packs, scam lists, the agency's own flags and partner agencies' flag packs, and classified: pass-through, exchange (an exit point), sanctioned, mixer, smart contract, or busy service. Bitcoin addresses spent together are grouped into clusters (a stated heuristic).
4. Results appear as a left-to-right money-flow map (victim on the left, exits on the right) with plain-language explanations on every element, plus ranked exit-point cards.
5. One click produces the evidence package: court-ready PDF report, chain-of-custody CSV, raw JSON, DRAFT affidavit methodology, per-exit traceroutes and DRAFT freeze letters with a SHA-256 manifest - plus an FBI IC3 complaint worksheet prefilled from the trace. Wallets can be watched for movement, flagged across cases, bulk-triaged, and checked against Chainabuse on demand.

Every single network request the tool makes is logged with its exact URL, UTC time and a SHA-256 hash of the response, so an independent analyst can re-run and verify any piece of the work. A separate Technical Validation Document describes the methodology in full for expert review.

Running the program

Requirements: Windows 10/11 (installed build) and a modern browser; from source, any OS with Python 3.10 or newer. Internet access for live tracing (previously fetched records are served from the local evidence cache).

Install (once): run the setup program (CryptoInvestigator-Setup-v<version>.exe; no administrator rights needed) or unzip the portable build. From source: `python -m pip install -r requirements.txt`.

Start: open Crypto Investigator from the Start menu or desktop shortcut (from source: `python run.py`). The browser opens at `http://127.0.0.1:8321`; an icon in the notification area shows the program is running and offers Open and Quit. No console window appears. The app runs entirely on the local machine; nothing is exposed to the network and case data never leaves the computer.

Where data lives: the installed build keeps its database, reports, logs and editable label files under %LOCALAPPDATA%\CryptoInvestigator\data (shown in Settings); a source checkout uses its own data/ folder. Either can be redirected (e.g. to an encrypted volume) with a data-location.txt file next to the program.

First-time setup (Settings button): no accounts are required. Run the four label downloads once (OFAC sanctions list, exchange label packs, scam blacklist, Ethereum name tags); they refresh themselves afterwards. Fill in the agency letterhead if freeze requests will be generated. Optional free API keys only raise rate limits.

Workflow: Step 1 create/pick a case → Step 2 paste the victim's wallet address → Step 3 (optional) paste the transaction hash of the payment to investigate → Step 4 choose how many hops to follow → press 'Follow the money' → review the exit points and map → download the report, custody log and IC3 worksheet.

What comes out

- Court-ready fund-tracing report (PDF) with methodology, limitations and chain-of-custody appendices.
- Chain-of-custody log (CSV) - every data pull, re-runnable.
- Raw trace data (JSON) for archival or other tools.
- FBI IC3 complaint worksheet (PDF) mirroring the official form at complaint.ic3.gov, prefilled with the traced payments.
- DRAFT freeze / preservation request letters, affidavit methodology draft, per-exit traceroutes, map image, evidence ZIP with SHA-256 manifest, bulk-triage CSV, flag packs for partner agencies.

Where to look next

- **Technical Validation Document** ([docs/validation/](#)) - full methodology, integrity controls, and step-by-step verification procedures for outside experts.
- **CHANGELOG.md** - what each version added and its known limitations. **docs/listings/** - the complete source code of every version in a single hashed file.